Atomic Lab: A Technical Whitepaper on Agentic RAG and Atomic Knowledge Workflows

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Abstract

Welcome to Atomic Lab. This document serves as both a technical whitepaper and an interactive demo file for our system. The exponential growth of scientific literature presents a dual crisis: the fragmentation of reading inspirations and the difficulty of tracing citations during writing. Atomic Lab solves this by deconstructing literature into "Atomic Knowledge" and leveraging an Agentic RAG (Retrieval-Augmented Generation) architecture. This manual outlines our core technical implementations and the empirical results from a 100-question academic QA benchmark.

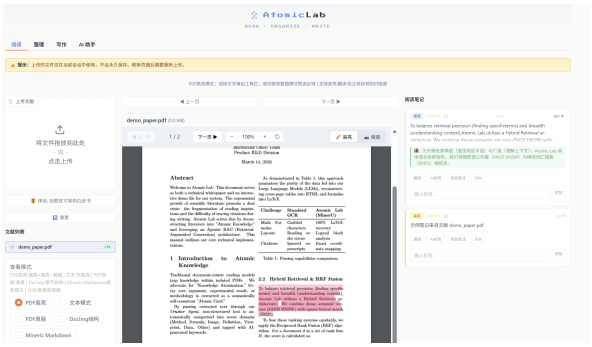


Figure 1: Atomic Lab

1 Introduction to Atomic Knowledge

Traditional document-centric reading models trap knowledge within isolated PDFs. We advocate for "Knowledge Atomization." Every core argument, experimental result, or methodology is extracted as a semantically self-consistent "Atomic Card."

By passing extracted text through our *note distiller* and *atomic decomposer* agents, non-structured text is automatically distilled into an axiom–method–boundary triple and tagged with AI-generated keywords.

2 Core Technologies

2.1 Multi-modal Deep Parsing

Academic literature parsing is notoriously difficult due to dual-column layouts and complex mathematical embeddings. Atomic Lab integrates the advanced **MinerU** parsing engine.

As demonstrated in Table 1, this approach guarantees the purity of the data fed into our Large Language Models (LLMs), reconstructing cross-page tables into HTML and formulas into LaTeX.

Challenge	**Standard OCR**	**Atomic Lab (MinerU)**
Math Formulas	Garbled characters	100% LaTeX recovery
Layouts	Reading order errors	Logical block analysis
Citations	Ignored superscripts	Exact coordinate mapping

Table 1: Parsing capabilities comparison.

2.2 Hybrid Retrieval & RRF Fusion

To balance retrieval precision (finding specific terms) and breadth (understanding context), Atomic Lab utilizes a Hybrid Retrieval architecture. We combine dense semantic vectors (via the embedded **ChromaDB** vector store) with sparse lexical search (BM25) and a knowledge graph.

To fuse these ranking systems equitably, we apply the Reciprocal Rank Fusion (RRF) algorithm. For a document d in a set of rank lists R, the score is calculated as:

$$RRF(d) = \sum_{r \in R} \frac{1}{k + r(d)} \quad (1)$$

where $k = 60$ acts as a smoothing constant. We use a *pure RRF* fusion (rank-only, no raw-score mixing), so that only knowledge atoms ranking high across multiple strategies are presented to the user.

An optional cross-encoder re-ranker further amplifies semantic relevance. Our ablation shows the re-ranking gain is *conditional on the BM25 weight*: on multi-hop questions a high BM25 weight (1.5) plus re-ranking yields +11.9 points, whereas on abstract comparative questions re-ranking hurts regardless of weight (see Table 2).

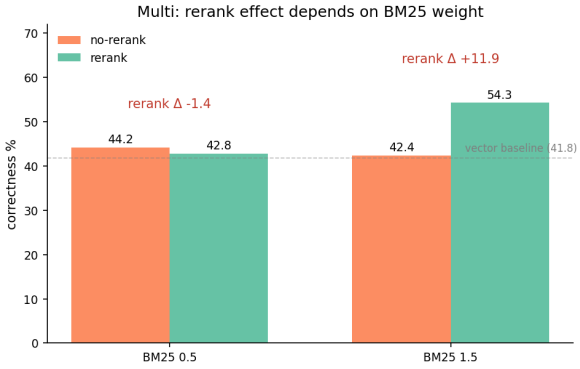


Figure 2: Re-ranking gain depends on the BM25 weight (multi-hop).

2.3 GraphRAG & 1/2-Hop Expansion

Knowledge in Atomic Lab is not flat. Atoms are stored in a dual-track architecture. When a user queries a concept, the system retrieves direct matches and performs a **1/2-hop expansion** along the knowledge graph (following shared tags or citation links) to uncover hidden academic causality.

3 The Agentic RAG Workflow

The chat interface of Atomic Lab is not a simple Q&A bot; it simulates a multi-agent "virtual research group" meeting of four roles:

- **Router:** Classifies user intent (academic retrieval / writing / paper recommendation / peer review) and dispatches to the appropriate tool via LLM function-calling.
- **Seeker:** Executes multi-source retrieval, pulling evidence from document vectors, BM25, note cards, the knowledge graph, and external fallback APIs (e.g., arXiv).
- **Reviewer:** Critically evaluates the retrieved context; if the evidence is insufficient, it flags low-confidence context and triggers re-retrieval.
- **Synthesizer:** Generates the final, logically coherent response, embedding precise, verifiable citations.

Ablating this orchestration (full vs. retrieval-only) on our benchmark shows the multi-agent chain adds +12.8 **points** on single-hop and +14.9 **points** on multi-hop correctness, while also reducing hallucination (single-hop 4% → 0%, multi-hop 36% → 20%).

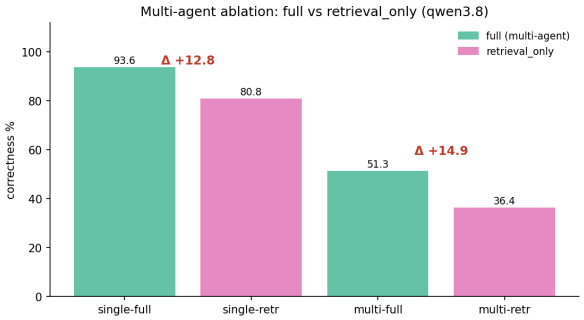


Figure 3: Multi-agent orchestration ablation (full vs. retrieval-only).

4 Evaluation Results

We evaluate on a self-built 100-question academic QA set (single-hop / multi-hop / comparative / adversarial, 25 each), judged by a fixed reasoning model with 95% confidence intervals.

Generation quality: faithfulness ≈ 93%, hallucination rate ≤ 8%. These results demonstrate that the hybrid retrieval pipeline delivers the largest gains on difficult multi-hop and comparative questions, where a single vector channel alone is insufficient.

Question type	Vector	Hybrid	Δ
Single-hop	86.0%	86.4%	+0.4
Multi-hop	41.8%	54.3%	+12.5
Comparative	37.8%	47.2%	+9.4
Adversarial	≈98%	≈98%	∼ 0
Overall	66.1%	71.4%	+5.3

Table 2: Correctness by question type (hybrid vs. vector-only).

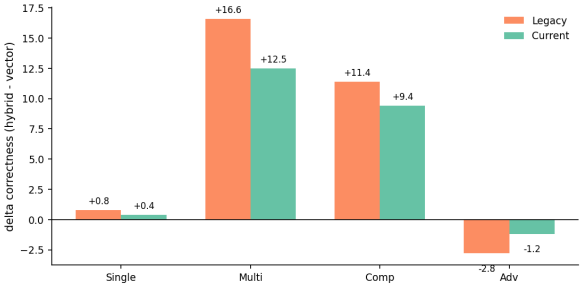


Figure 4: Per-type correctness improvement (hybrid vs. vector-only).

5 Conclusion

By integrating deep document parsing, hybrid retrieval (ChromaDB + BM25 + GraphRAG with pure RRF fusion), and a four-role Agentic workflow, Atomic Lab provides a seamless *Read–Organize–Write* loop. We invite you to explore the system by highlighting texts, asking questions, and generating your own verifiable academic insights.